SQL Data Types for Beginners

A beginner-friendly overview of SQL data types, examples, and whether constraints are commonly used with them.

1. What is a Data Type?

A data type tells SQL what kind of data a column can store, such as numbers, text, dates, or true/false values.

2. Common SQL Data Types

Data Type	Use	Example	Constraints?
INT	Whole numbers	CustomerID INT	Often: PRIMARY KEY, NOT NULL, UNIQUE
BIGINT	Very large whole numbers	SalesID BIGINT	Often yes
DECIMAL(p,s)	Exact decimals	Price DECIMAL(10,2)	Often yes
FLOAT	Approximate decimals	Score FLOAT	Optional
CHAR(n)	Fixed-length text	Code CHAR(5)	Optional
VARCHAR(n)	Variable-length text	Name VARCHAR(100)	Often yes
TEXT	Large text	Comments TEXT	Usually optional
DATE	Dates only	OrderDate DATE	Often yes
TIME	Time only	StartTime TIME	Optional
DATETIME	Date and time	CreatedDate DATETIME	Often yes
BOOLEAN/BIT	True or False	IsActive BIT	Often yes
BINARY/VARBINARY	Binary data	FileData VARBINARY(MAX)	Optional

3. Most Common Constraints

NOT NULL: Column cannot be empty

PRIMARY KEY: Uniquely identifies each row

FOREIGN KEY: Links tables together

UNIQUE: No duplicate values

DEFAULT: Sets a default value

CHECK: Controls allowed values

4. Example Table

```
CREATE TABLE Customers (
 CustomerID INT PRIMARY KEY,
 FullName VARCHAR(100) NOT NULL,
 CreditLimit DECIMAL(10,2),
 JoinDate DATE DEFAULT GETDATE()
);
```

5. Beginner Rules

- Use INT for whole numbers.
- Use DECIMAL for money.
- Use VARCHAR for names and text.
- Use DATE/DATETIME for dates.
- Use constraints to improve data quality.
- Not every column requires constraints.